UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	INTRODUCTION TO INTERACTIVE MULTIMEDIA
COURSE CODE	CSC413
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

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PART A (15 MARKS)

State TRUE (T) or FALSE (F) for the following statements.

1. Authoring tools help developers organise and edit multimedia elements.
2. 3D animation does not use modelling.
3. A storyboard is a visual plan of a multimedia sequence.
4. Hertz measures sound frequency.
5. A vector image becomes pixelated when enlarged.
6. MPEG is a common family of video compression formats.
7. Anti-aliasing makes the edges of text look more jagged.
8. CMYK is used for printed output.
9. MIDI files store digitized audio samples.
10. A plotter is an input device.
11. Hypermedia combines multimedia with hyperlinks.
12. A web browser displays hypertext documents.
13. USB is a plug-and-play connection standard.
14. RGB is an additive colour model.
15. Interlacing displays every line of a video frame at the same time.

PART B (15 MARKS)

Select the most appropriate answer from Table A for each of the following questions.

Table A. Matching options

A. Frame rate	B. HTML	C. Browser
D. Linear multimedia	E. Hypermedia	F. Vector
G. Scanner	H. Copyright	I. Bitmap
J. Multimedia	K. Card-based tool	L. Keyframe
M. Virtual reality	N. Tweening	O. Storyboard

1. integration of text, graphics, audio, video and animation
2. simulated environment
3. frames displayed per second
4. image made of pixels
5. visual sequence plan
6. important frame in an animation
7. software that displays web pages
8. input device that digitises documents
9. presentation without user control of sequence
10. image made of mathematical objects
11. linked multimedia elements
12. language for web structure
13. exclusive rights of a creator
14. generating in-between frames
15. authoring tool organised like pages or cards

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PART C (30 MARKS)

Choose the correct answer for each question.

1. A wireframe is:
   A. A final video
   B. A sound file
   C. A low-fidelity layout sketch
   D. A codec

2. Which is an authoring tool category?
   A. Card-based
   B. CMYK
   C. RGB
   D. JPEG

3. Hypertext contains:
   A. Only images
   B. Links to other content
   C. No text
   D. Only audio

4. Which is an audio file format?
   A. JPEG
   B. MPEG
   C. WAV
   D. HTML

5. Which is NOT a multimedia element?
   A. Text
   B. Audio
   C. Compiler
   D. Animation

6. HTML is used to:
   A. Store audio samples
   B. Compress video
   C. Structure web content
   D. Model 3D objects

7. Which best defines multimedia?
   A. A single image
   B. Text, graphics, audio, video and animation
   C. Text only
   D. A printed book

8. Morphing changes:
   A. One shape into another
   B. Audio pitch
   C. File format
   D. Font size

9. RAM is:
   A. Permanent storage
   B. Temporary working memory
   C. An input device
   D. A web browser

10. Frame rate is:
    A. Bits per second
    B. Samples per second
    C. Frames per second
    D. Pixels per inch

11. Which format is commonly used for bitmap images?
    A. MIDI
    B. PNG
    C. HTML
    D. MPEG

12. Tweening creates:
    A. In-between frames
    B. Text fonts
    C. Video codecs
    D. Sound effects

13. A prototype is:
    A. A font
    B. An early model of a product
    C. A final release
    D. A codec

14. Interlacing:
A. Draws alternate lines of a frame
B. Removes colour
C. Draws every line at once
D. Adds sound

15. Which is a benefit of multimedia in education?
A. It removes all interaction
B. It replaces all teachers
C. It uses only text
D. It combines visuals and sound

16. RGB is used mainly for:
A. Audio recording
B. Printed output
C. Screen display
D. Video editing only

17. Which is an input device?
A. Scanner
B. Printer
C. Monitor
D. Plotter

18. A presentation where the user controls the sequence is called:
A. Non-linear
B. Static
C. Printed
D. Linear

19. Sampling converts:
A. Digital video to text
B. Images to vectors
C. Text to speech
D. Analog sound to digital values

20. Which is a 3D animation task?
A. Bookmarking
B. Modelling and rendering
C. Kerning
D. Sampling

21. A serif font has:
A. No strokes
B. Small strokes at character ends
C. Only uppercase letters
D. No spacing

22. Keyframes are:
A. Colour profiles
B. Web links
C. Important frames in an animation
D. Audio samples

23. Card-based authoring tools are suited to:
A. Live video only
B. Messages with a beginning and end
C. 3D modelling only
D. Audio editing only

24. Which colour model is subtractive?
A. HSV
B. CMYK
C. RGB
D. HSL

25. Hertz measures:
A. Loudness
B. Resolution
C. Frequency
D. File size

26. Which is an example of linear multimedia?
A. A game with choices
B. An interactive kiosk
C. A cinema presentation
D. A website with menus

27. Amplitude of a sound wave relates to:
A. Pitch
B. Loudness
C. Colour
D. Frame rate

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28. A hyperlink links to:

A. Another resource
B. Only text
C. Only audio
D. Only images

29. Convergence in multimedia refers to:

A. Merging of media and computing
B. Deleting media files
C. Recording audio only
D. Printing images

30. Plug and play means:

A. Manual driver installation only
B. No power needed
C. Wireless only
D. Automatic device recognition

PART D (40 MARKS)

QUESTION 1

Text is an essential element in a multimedia project. Its shape, size, design, colour, arrangement and linking all affect the message.

a) State TWO reasons sans serif is mostly used for screen display and serif is used for printed material.

(3 marks)

b) Describe TWO roles of hypertext in multimedia.

(3 marks)

c) Explain how text can have a massive impact on an informative multimedia project in the link and HTML context.

(4 marks)

QUESTION 2

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

(2 marks)

b) Explain the interactivity level suitable for the demonstration.

(3 marks)

c) Explain TWO authoring tool categories appropriate for this project.

(5 marks)

QUESTION 3

A museum wants a website that presents its collection and lets visitors explore related exhibits.

a) State TWO differences between linear and non-linear navigation.

(2 marks)

b) Describe TWO roles of hyperlinks in the website.

(4 marks)

c) Explain how streaming and compression affect multimedia delivery on the web.

(4 marks)

QUESTION 4

A shopping complex requires a kiosk that gives customers location and retailer information.

a) State TWO important project team roles for completing the kiosk.

(3 marks)

b) Design a low-fidelity prototype for a pre-school courseware home page. The page must show topic menus, the course title and the school logo.

(3 marks)

c) Discuss TWO factors that lead to the wide use of mobile devices in games and shopping applications.

(4 marks)

END OF QUESTION PAPER

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